

Test II

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Trigonometry).
- II) Show all work.
- III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You will need a calculator for this test (be sure that it is in degree mode).
- V) Each problem is worth four points for a total of 100 points.

Prove that the following equations are true by converting the left side to the right.

$$\textcircled{1} \frac{\cos(2B)}{\sin B} = \csc B - 2\sin B$$

$$\textcircled{2} \sin^3 \theta + \sin \theta \cos^2 \theta = \sin \theta$$

$$\textcircled{3} \frac{\tan \theta}{\cot \theta} = \tan^2 \theta$$

$$\textcircled{4} \left(\frac{1}{\cos w}\right)^2 + 36 = \tan^2 w + 37$$

$$\textcircled{5} \sin(2\beta) - \sin^2 \beta \frac{1}{\tan \beta} = \sin \beta \cos \beta$$

$$\textcircled{6} \tan x + \frac{\cos x}{1 + \sin x} = \sec x$$

$$\textcircled{7} \cos(x+y) - \cos(x-y) = -2\sin x \sin y$$

$$\textcircled{8} \frac{\sec^2 A - \csc^2 A}{\sec A + \csc A} = \sec A + \csc(A + 180^\circ)$$

Simplify the expressions.

$$\textcircled{9} \cos(-\alpha) + \frac{1}{\sec \alpha} + \sin(\alpha - 90^\circ)$$

$$\textcircled{10} \cos(10x) \cos(8x) + \sin(10x) \sin(8x)$$

$\textcircled{11}$ Find the value of the expression below **without a calculator.**

$$\sin 15^\circ$$

⑫ Find all values of θ .

$$5\sqrt{3}\sec\theta + 6 - 3\sqrt{3}\sec\theta = \sqrt{3}\sec\theta + 8$$

⑬ Simplify the expression below.

$$\cot^2(360^\circ - B) + 1$$

⑭ Solve if $0 < W \leq 360^\circ$.

$$3\csc W - 4\sin W = 0$$

⑮ Solve if $0 < B < 360^\circ$.

$$\cos\left(\frac{B}{2} - 270^\circ\right) = -1$$

⑯ Solve if $0 \leq A < 360^\circ$. Round to the nearest tenth.

$$\sin A + 1 = \csc A$$

⑪ Solve if $0 < \beta \leq 360^\circ$.

$$2\csc(3\beta) + 2 = 4\csc(3\beta)$$

⑫ Solve if $0 \leq \alpha < 360^\circ$. ⑬ Solve if $0 < A < 360^\circ$.

$$9\cos^2\alpha - 4 - \sin^2\alpha = \cos(2\alpha) \quad \cot A = 2\cos A$$

Prove that the equations are true by converting the left side to the right.

$$(20) \frac{\cot^4 x + \tan^2 x}{\tan x \cot^4 x} = \frac{\cos x}{\sin x} + \tan^5 x$$

$$(21) \frac{\cot B + 5}{6 \cot^2 B + 25 \cot B - 25} = \frac{1}{6 \cot B - 5}$$

② Solve if $0 < A < 360^\circ$. Round to the nearest thousandth if necessary.

$$14 + 3\tan^2 A = 4\sec^2 A - 9\tan A$$

③ If W is in standard position, and it terminates in QII ($90^\circ < W < 180^\circ$), find the value of each expression below if $\cos W = -\frac{2}{5}$.

i) $\sin\left(\frac{W}{2}\right)$

ii) $\tan\left(\frac{W}{2}\right)$

24) If B is in standard position, and it terminates in QII ($90^\circ < B < 180^\circ$), find the value of the expression below if $\cos B = -\frac{1}{3}$.

i) $\cos(2B)$

ii) Find the value of $\sin(2B)$ using the double angle formula below.

$$\sin(2B) = 2\sin B \cos B$$

Q5) If β is in standard position, and it terminates in QIV ($270^\circ < \beta < 360^\circ$), find the value of each expression below if $\sin \beta = -\frac{\sqrt{11}}{6}$.

i) $\cos\left(\frac{\beta}{2}\right)$

ii) $\csc\left(\frac{\beta}{2}\right)$

Test II Answers

$$\textcircled{1} \frac{\cos(2\beta)}{\sin\beta} = \csc\beta - 2\sin\beta \quad \textcircled{2} \sin^3\theta + \sin\theta\cos^2\theta = \sin\theta$$

$$\frac{1 - 2\sin^2\beta}{\sin\beta} =$$

$$\frac{1}{\sin\beta} - \frac{2\sin^2\beta}{\sin\beta} =$$

$$\csc\beta - 2\sin\beta =$$



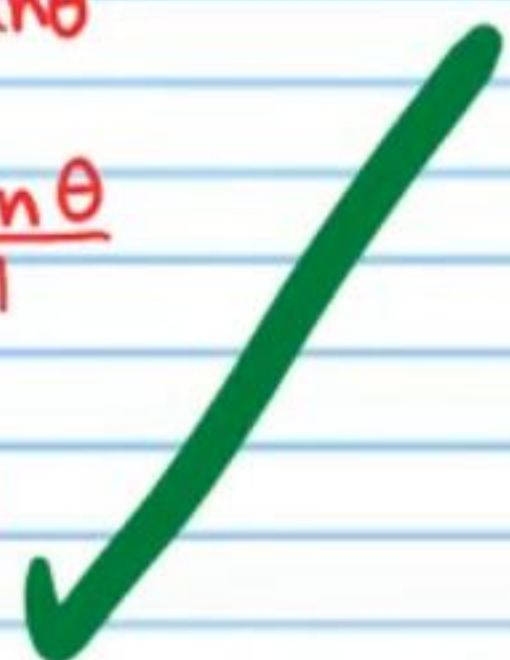
$$\textcircled{3} \frac{\tan\theta}{\cot\theta} = \tan^2\theta$$

$$\frac{\tan\theta}{\frac{1}{\tan\theta}} =$$

$$\tan\theta \div \frac{1}{\tan\theta}$$

$$\tan\theta \cdot \frac{\tan\theta}{1}$$

$$\tan^2\theta$$



$$\sin\theta(\sin^2\theta + \cos^2\theta) =$$

$$\sin\theta(1) =$$

$$\sin\theta =$$



$$\textcircled{4} \left(\frac{1}{\cos w}\right)^2 + 36 = \tan^2 w + 37$$

$$(\sec w)^2 + 36 =$$

$$\sec^2 w + 36 =$$

$$\tan^2 w + 1 + 36 =$$

$$\tan^2 w + 37 =$$



$$\textcircled{5} \sin(2\beta) - \sin^2\beta \frac{1}{\tan\beta} = \sin\beta \cos\beta$$

$$2\sin\beta \cos\beta - \sin^2\beta \frac{1}{\tan\beta} =$$

$$2\sin\beta \cos\beta - \sin^2\beta \cot\beta =$$

$$2\sin\beta \cos\beta - \sin^2\beta \frac{\cos\beta}{\sin\beta} =$$

$$2\sin\beta \cos\beta - \sin\beta \cos\beta =$$

$$\sin\beta \cos\beta =$$



$$\textcircled{6} \tan x + \frac{\cos x}{1 + \sin x} = \sec x$$

$$\frac{\sin x}{\cos x} + \frac{\cos x}{1 + \sin x} =$$

$$\frac{\sin x + \sin^2 x}{\cos x(1 + \sin x)} + \frac{\cos^2 x}{\cos x(1 + \sin x)} =$$

$$\frac{\sin x + \sin^2 x + \cos^2 x}{\cos x(1 + \sin x)} =$$

$$\frac{\sin x + 1}{\cos x(1 + \sin x)} =$$

$$\frac{1 + \sin x}{\cos x(1 + \sin x)} =$$

$$\frac{1}{\cos x} =$$

$$\sec x =$$



$$\textcircled{7} \cos(x+y) - \cos(x-y) = -2\sin x \sin y$$

$$\cos x \cos y - \sin x \sin y - [\cos x \cos y + \sin x \sin y] =$$

$$\cos x \cos y - \sin x \sin y - \cos x \cos y - \sin x \sin y =$$

$$-2\sin x \sin y =$$



$$\textcircled{8} \frac{\sec^2 A - \csc^2 A}{\sec A + \csc A} = \sec A + \csc(A + 180^\circ)$$

$$\frac{(\sec A + \csc A)(\sec A - \csc A)}{\sec A + \csc A} =$$

$$\sec A - \csc A =$$

$$\sec A + \csc(A + 180^\circ) =$$



$$(9) \cos \alpha$$

$$(10) \cos(2x)$$

$$(11) \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$(12) \begin{aligned} \theta &= 30^\circ + 360n, n \text{ is any integer.} \\ \theta &= 330^\circ + 360n, n \text{ is any integer.} \end{aligned}$$

$$(13) \csc^2 B$$

$$(14) W = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$

$$(15) B = 180^\circ$$

$$(16) A \approx 38.2^\circ, 141.8^\circ$$

$$(17) \beta = 30^\circ, 150^\circ, 270^\circ$$

$$(18) \alpha = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

$$(19) A = 30^\circ, 90^\circ, 150^\circ, 270^\circ$$

$$(20) \frac{\cot^4 x + \tan^2 x}{\tan x \cot^4 x} = \frac{\cos x}{\sin x} + \tan^5 x$$

$$\frac{\cot^4 x}{\tan x \cot^4 x} + \frac{\tan^2 x}{\tan x \cot^4 x} =$$

$$\frac{1}{\tan x} + \frac{\tan x}{\cot^4 x} =$$

$$\frac{1}{\tan x} + \tan x \frac{1}{\cot^4 x} =$$

$$\cot x + \tan x (\tan^4 x) =$$

$$\cot x + \tan^5 x =$$

$$\frac{\cos x}{\sin x} + \tan^5 x =$$



$$(21) \frac{\cot B + 5}{6\cot^2 B + 25\cot B - 25} = \frac{1}{6\cot B - 5}$$

$$\frac{\cot B + 5}{(6\cot B - 5)(\cot B + 5)} =$$

$$\frac{1}{6\cot B - 5} =$$



$$(22) \boxed{A = 84.289^\circ, 135^\circ, 264.289^\circ, 315^\circ}$$

$$(23) \text{ i) } \boxed{\frac{\sqrt{70}}{10}} \quad \text{ii) } \frac{\sqrt{70}}{\sqrt{30}} = \frac{\sqrt{2100}}{30} = \frac{10\sqrt{21}}{30} = \boxed{\frac{\sqrt{21}}{3}}$$

$$(24) \text{ i) } \boxed{-\frac{7}{9}}$$

$$\text{ii) } \sin(2B) = 2\sin B \cos B = 2\left(\frac{2\sqrt{2}}{3}\right)\left(-\frac{1}{3}\right) = \boxed{-\frac{4\sqrt{2}}{9}}$$

$$(25) \text{ i) } \boxed{-\frac{\sqrt{33}}{6}}$$

$$\text{ii) } \frac{6}{\sqrt{3}} = \boxed{2\sqrt{3}}$$